A polynomial with no root in the disc

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Let n be a positive integer, and let

$$f_n(z) = n + (n-1)z + (n-2)z^2 + \cdots + z^{n-1}.$$

Prove that f_n has no roots in the closed unit disc $\{z \in \mathbb{C} : |z| \leq 1\}$.

Proof. Suppose, for a contradiction, that there exists $z \in \mathbb{C}$ such that $|z| \leq 1$ and

$$\sum_{k=0}^{n-1} (n-k) z^k = 0,$$

that is, $f_n(z) = 0$.

For $m \geq 0$ and $w \in \mathbb{C}$, write

$$S_m(w) = \sum_{k=0}^{m-1} (m-k) w^k,$$

with the empty sum $S_0(w) = 0$, so that $f_n(z) = S_n(z)$.

First, we prove the geometric sum identity: for all m and w,

$$(1-w) \sum_{k=0}^{m-1} w^k = 1 - w^m. \tag{1}$$

This is clear for $m = 0$. If it holds for m, then

$$(1-w) \sum_{k=0}^{m} w^k = (1-w) \left(\sum_{k=0}^{m-1} w^k + w^m \right) = (1-w^m) + (1-w) w^m = 1 - w^{m+1},$$

which is the identity for $m+1$.

Next, for every m and w,

$$S_{m+1}(w) = S_m(w) + \sum_{k=0}^{m} w^k. \tag{2}$$

Indeed, for $0 \leq k \leq m$ we have $(m+1-k) = (m-k)+1$, hence $(m+1-k)w^k = (m-k)w^k + w^k$. Summing over $0 \leq k \leq m$ gives

$$S_{m+1}(w) = \sum_{k=0}^{m} (m-k) w^k + \sum_{k=0}^{m} w^k,$$

and the first sum on the right is $S_m(w)$, since its term with $k = m$ is 0.

We now prove the key identity: for all m and w,

$$(1-w) S_m(w) = m - \sum_{k=0}^{m-1} w^{k+1}. \tag{3}$$

For $m = 0$ both sides are 0. Assuming (3) for m, and using (2), (1), we obtain

$$\begin{aligned}(1-w)S_{m+1}(w) &= (1-w)S_m(w) + (1-w)\sum_{k=0}^{m} w^k \\ &= \left(m - \sum_{k=0}^{m-1} w^{k+1}\right) + \left(1 - w^{m+1}\right) \\ &= (m+1) - \sum_{k=0}^{m} w^{k+1},\end{aligned}$$

which is (3) with m replaced by $m+1$.

Apply (3) with $m = n$ and $w = z$. Since $S_n(z) = f_n(z) = 0$, the left-hand side is 0, so

$$n = \sum_{k=0}^{n-1} z^{k+1} \quad \text{in } \mathbb{C}. \tag{4}$$

Taking real parts in (4), and using that the real part of a sum is the sum of the real parts, gives

$$n = \sum_{k=0}^{n-1} \mathrm{Re}\left(z^{k+1}\right) \quad \text{in } \mathbb{R}. \tag{5}$$

For each k with $0 \le k \le n-1$,

$$\mathrm{Re}\left(z^{k+1}\right) \le \left|z^{k+1}\right| = |z|^{k+1} \le 1^{k+1} = 1,$$

using $|z| \le 1$. Hence every term $1 - \mathrm{Re}(z^{k+1})$ is nonnegative. But by (5),

$$\sum_{k=0}^{n-1} \left(1 - \mathrm{Re}\left(z^{k+1}\right)\right) = n - \sum_{k=0}^{n-1} \mathrm{Re}\left(z^{k+1}\right) = 0.$$

A sum of nonnegative real numbers being 0, each term is 0:

$$1 - \mathrm{Re}\left(z^{k+1}\right) = 0 \qquad (0 \le k \le n-1).$$

Taking $k = 0$, which is allowed since $n > 0$, gives

$$\mathrm{Re}(z) = 1.$$

Moreover, $|z|^2 = \mathrm{Re}(z)^2 + \mathrm{Im}(z)^2 \le 1$. Substituting $\mathrm{Re}(z) = 1$ gives $1 + \mathrm{Im}(z)^2 \le 1$, so $\mathrm{Im}(z)^2 = 0$ and therefore $\mathrm{Im}(z) = 0$. Thus $z = 1$.

But then

$$0 = f_n(z) = f_n(1) = \sum_{k=0}^{n-1} (n-k).$$

This is impossible: the sum on the right is a sum of nonnegative integers whose term with $k = 0$ is $n > 0$, so it is strictly positive, in particular nonzero as a complex number. This contradiction shows that f_n has no root with $|z| \le 1$. $\square$